

Pegged Ahmed-Body Comparison to ERCOFTAC-Style Benchmarks

Geodesic Sectional Boundary Method with Morphed Normal Shell Diagnostics

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Purpose

This document compares a pegged Ahmed-body computational result against ERCOFTAC-style Ahmed-body benchmark expectations. The important correction is that the body is not a clean Ahmed body. It contains pegs or stilts. Therefore, the comparison must include peg drag, underbody disturbance, support-body interaction, and solver error.

The result should not be judged only against a clean Ahmed-body drag coefficient.

Reported Harding Run

The uploaded Android-local CFD report gives the following force diagnostics:

$$C_{D,\text{wake}} = 0.376743,$$

$$C_{D,\text{geoP}} = 0.407006.$$

Thus the practical drag range is

$$C_D \approx 0.38\text{--}0.41.$$

The run is best described as

Android real-Re RANS/DDES-style geodesic sectional CFD.

It is not DNS. The wake-momentum value is the main drag estimate, while the geodesic-pressure value is a boundary-pressure diagnostic that still requires workstation validation.

Clean Ahmed-Body Benchmark Target

A common clean Ahmed-body benchmark value for the 25-degree slant configuration is approximately

$$C_{D,\text{clean}} \approx 0.285.$$

Some validation sources report clean Ahmed-body values closer to

$$C_D \approx 0.285\text{--}0.299.$$

Therefore a practical clean benchmark band is

$$C_{D,\text{clean}} \approx 0.285\text{--}0.299.$$

However, this clean-body band is not the correct final comparison if the tested body has pegs or stilts.

Correct Drag Decomposition for the Pegged Body

For a pegged or stilted Ahmed body, the correct drag model is

$$C_D^{\text{your}} = C_D^{\text{clean}} + \Delta C_D^{\text{pegs}} + \Delta C_D^{\text{underbody}} + \Delta C_D^{\text{interaction}} + \Delta C_D^{\text{solver}}.$$

Here,

$$C_D^{\text{clean}}$$

is the clean Ahmed-body drag,

$$\Delta C_D^{\text{pegs}}$$

is direct drag from the pegs,

$$\Delta C_D^{\text{underbody}}$$

is extra drag caused by disturbed underbody flow,

$$\Delta C_D^{\text{interaction}}$$

is peg-body-wake interference,

and

$$\Delta C_D^{\text{solver}}$$

is numerical error from grid coarseness, pressure projection, turbulence modeling, and convergence.

Thus the correct comparison is not

$$C_D^{\text{your}} \quad \text{versus} \quad C_D^{\text{clean}}.$$

The correct comparison is

$$C_D^{\text{your}} \quad \text{versus} \quad C_D^{\text{benchmark+pegs}}.$$

Difference from the Clean Benchmark

Using

$$C_{D,\text{clean}} = 0.285,$$

the wake-momentum estimate differs by

$$\Delta C_{D,\text{wake}} = 0.376743 - 0.285.$$

Therefore,

$$\Delta C_{D,\text{wake}} = 0.091743.$$

The relative difference is

$$\frac{0.091743}{0.285} = 0.3219,$$

or

$$32.2\%.$$

For the geodesic-pressure diagnostic,

$$\Delta C_{D,\text{geoP}} = 0.407006 - 0.285.$$

Therefore,

$$\Delta C_{D,\text{geoP}} = 0.122006.$$

The relative difference is

$$\frac{0.122006}{0.285} = 0.4281,$$

or

$$42.8\%.$$

Without pegs, these numbers would look too high. With pegs, these differences are not pure errors. Instead,

$$0.091743 = \Delta C_D^{\text{pegs}} + \Delta C_D^{\text{underbody}} + \Delta C_D^{\text{interaction}} + \Delta C_D^{\text{solver,wake}},$$

and

$$0.122006 = \Delta C_D^{\text{pegs}} + \Delta C_D^{\text{underbody}} + \Delta C_D^{\text{interaction}} + \Delta C_D^{\text{solver,geoP}}.$$

Peg Projected-Area Estimate

A first-order peg drag correction may be estimated by projected area:

$$\Delta C_D^{\text{pegs}} \approx \frac{1}{A_{\text{ref}}} \sum_{k=1}^{N_p} C_{D,k}^{\text{peg}} A_{p,k} \eta_k.$$

For a cylindrical peg,

$$A_{p,k} \approx d_k h_k.$$

Therefore,

$$\Delta C_D^{\text{pegs}} \approx \frac{1}{A_{\text{ref}}} \sum_{k=1}^{N_p} C_{D,k}^{\text{peg}} d_k h_k \eta_k.$$

The variables are:

$$N_p = \text{number of pegs},$$

$$d_k = \text{peg diameter},$$

$$h_k = \text{exposed peg height},$$

$$C_{D,k}^{\text{peg}} = \text{isolated peg drag coefficient},$$

$$\eta_k = \text{shielding or interference factor},$$

$$A_{\text{ref}} = \text{Ahmed-body frontal reference area}.$$

For four pegs,

$$N_p = 4.$$

If the support dimensions are approximately

$$d = 0.03 \text{ m},$$

$$h = 0.05 \text{ m},$$

then the total peg projected area is

$$A_{\text{pegs}} = N_p dh.$$

Substituting,

$$A_{\text{pegs}} = 4(0.03)(0.05).$$

Thus,

$$A_{\text{pegs}} = 0.006 \text{ m}^2.$$

Using an Ahmed-body frontal reference area,

$$A_{\text{ref}} \approx 0.389 \times 0.288,$$

so

$$A_{\text{ref}} \approx 0.112032 \text{ m}^2.$$

The peg projected-area ratio is therefore

$$\frac{A_{\text{pegs}}}{A_{\text{ref}}} = \frac{0.006}{0.112032}.$$

Thus,

$$\frac{A_{\text{pegs}}}{A_{\text{ref}}} = 0.0536.$$

The pegs alone therefore represent about

$$5.36\%$$

of the frontal reference area.

If

$$C_D^{\text{peg}} \approx 1.0\text{--}1.2$$

and

$$\eta \approx 0.5\text{--}1.0,$$

then

$$\Delta C_D^{\text{pegs}} \approx (1.0\text{--}1.2)(0.0536)(0.5\text{--}1.0).$$

Therefore,

$$\Delta C_D^{\text{pegs}} \approx 0.027\text{--}0.064.$$

Clean Body Plus Direct Peg Drag

If the clean Ahmed-body benchmark band is

$$C_{D,\text{clean}} \approx 0.285\text{--}0.299,$$

and the direct peg correction is

$$\Delta C_D^{\text{pegs}} \approx 0.027\text{--}0.064,$$

then a first-order pegged estimate becomes

$$C_D^{\text{clean+direct pegs}} \approx 0.312\text{--}0.363.$$

This estimate still does not include all effects. It excludes underbody vortex interaction, peg-body interference, wake modification, exact OBJ differences, and solver error.

Corrected Difference for the Wake-Momentum Estimate

The wake-momentum drag is

$$C_{D,\text{wake}} = 0.376743.$$

Using the first-order pegged estimate,

$$C_D^{\text{pegged}} \approx 0.312\text{--}0.363.$$

At the high end,

$$0.376743 - 0.363 = 0.013743.$$

At the low end,

$$0.376743 - 0.312 = 0.064743.$$

Therefore,

$$\Delta C_D^{\text{remaining,wake}} \approx 0.014\text{--}0.065.$$

As percentages,

$$\frac{0.013743}{0.363} = 0.0379,$$

and

$$\frac{0.064743}{0.312} = 0.2075.$$

Therefore the wake-momentum drag is approximately

$$3.8\%\text{--}20.8\%$$

above a simple pegged estimate.

This is much less severe than the direct clean-body comparison, which gave

$$32.2\%.$$

Corrected Difference for the Geodesic-Pressure Diagnostic

The geodesic-pressure drag is

$$C_{D,\text{geoP}} = 0.407006.$$

Using

$$C_D^{\text{pegged}} \approx 0.312\text{--}0.363,$$

the remaining difference is

$$0.407006 - 0.363 = 0.044006,$$

and

$$0.407006 - 0.312 = 0.095006.$$

Thus,

$$\Delta C_D^{\text{remaining, geoP}} \approx 0.044\text{--}0.095.$$

As percentages,

$$\frac{0.044006}{0.363} = 0.1212,$$

and

$$\frac{0.095006}{0.312} = 0.3045.$$

Therefore the geodesic-pressure diagnostic is approximately

$$12.1\%\text{--}30.5\%$$

above a simple pegged estimate.

This is plausible for a pressure diagnostic because the uploaded run reports that pressure projection and divergence convergence remain limitations.

Wake-Likelihood Correction for Pegs

The wake-likelihood model should include peg effects because pegs introduce local separation, lower-body vortices, and underbody disturbance.

A dimensionless wake-likelihood model with peg influence is

$$W(\mathbf{x}) = \sigma \left[\beta_0 + \beta_1 (1 - \hat{\mathbf{n}} \cdot \hat{\mathbf{u}}_\infty) + \beta_2 \kappa L_{\text{ref}} + \beta_3 \frac{L_{\text{ref}} |\nabla p|}{q_\infty} + \beta_4 \chi_{\text{peg}}(\mathbf{x}) \right].$$

The dynamic pressure is

$$q_\infty = \frac{1}{2} \rho U_\infty^2.$$

The peg-proximity or peg-wake indicator can be written as

$$\chi_{\text{peg}}(\mathbf{x}) = \sum_{k=1}^{N_p} \exp \left(-\frac{\|\mathbf{x} - \mathbf{x}_{p,k}\|^2}{\ell_p^2} \right) (1 - \hat{\mathbf{n}}_{p,k} \cdot \hat{\mathbf{u}}_\infty).$$

Here,

$$\mathbf{x}_{p,k}$$

is the location of peg k ,

$$\ell_p$$

is a peg-wake influence length scale,

and

$$\hat{\mathbf{n}}_{p,k}$$

is a representative peg-surface normal.

Each term inside the wake-likelihood brackets is dimensionless:

$$1 - \hat{\mathbf{n}} \cdot \hat{\mathbf{u}}_\infty \quad \text{is dimensionless,}$$

$$\kappa L_{\text{ref}} \quad \text{is dimensionless,}$$

$$\frac{L_{\text{ref}} |\nabla p|}{q_\infty} \quad \text{is dimensionless,}$$

and

$$\chi_{\text{peg}}(\mathbf{x}) \quad \text{is dimensionless.}$$

Therefore,

$$[\beta_0] = [\beta_1] = [\beta_2] = [\beta_3] = [\beta_4] = 1.$$

Correct Validation Error

The wrong clean-body validation error is

$$\epsilon_D = \frac{C_D^{\text{your}} - C_D^{\text{clean}}}{C_D^{\text{clean}}}.$$

That comparison is incomplete because the tested geometry includes pegs.

The correct matched validation error is

$$\epsilon_D^{\text{matched}} = \frac{C_D^{\text{your}} - C_D^{\text{benchmark+pegs}}}{C_D^{\text{benchmark+pegs}}}.$$

The pegged benchmark should be modeled as

$$C_D^{\text{benchmark+pegs}} = C_D^{\text{clean}} + \Delta C_D^{\text{pegs}} + \Delta C_D^{\text{underbody}} + \Delta C_D^{\text{support interaction}}.$$

Therefore, for the wake-momentum estimate,

$$\epsilon_{D,\text{wake}}^{\text{matched}} = \frac{0.376743 - C_D^{\text{benchmark+pegs}}}{C_D^{\text{benchmark+pegs}}}.$$

For the geodesic-pressure diagnostic,

$$\epsilon_{D,\text{geoP}}^{\text{matched}} = \frac{0.407006 - C_D^{\text{benchmark+pegs}}}{C_D^{\text{benchmark+pegs}}}.$$

Comparison Table

Quantity	Clean Ahmed Body	Pegged Ahmed Body	Harding Run
Geometry	Clean slanted Ahmed body	Ahmed body plus pegs or stilts	OBJ Ahmed-body variant with pegs
Expected drag	$C_D \approx 0.285\text{--}0.299$	$C_D \approx 0.312\text{--}0.363$	$C_D \approx 0.38\text{--}0.41$
Main added physics	Rear separation and base pressure drag	from direct peg correction only Rear separation plus peg drag, underbody vortices, and support interaction	Wake-momentum and geodesic-pressure diagnostics
Comparison meaning	Too harsh if used alone	Better first-order comparison	Promising diagnostic, not final validation
Main limitation	Does not include pegs	Still simplified unless exact peg geometry is matched	Pressure projection and divergence convergence still need improvement

Interpretation

The old clean-body comparison was

$$C_D^{\text{your}} \approx 0.38\text{--}0.41$$

versus

$$C_D^{\text{clean}} \approx 0.285\text{--}0.299.$$

That made the result look too high.

The corrected pegged-body comparison is

$$C_D^{\text{your}} \approx 0.38\text{--}0.41$$

versus

$$C_D^{\text{clean}} + \Delta C_D^{\text{pegs}} + \Delta C_D^{\text{underbody}} + \Delta C_D^{\text{interaction}}.$$

With a first-order peg-area correction,

$$C_D^{\text{pegged}} \approx 0.31\text{--}0.36.$$

Therefore the wake-momentum estimate,

$$C_{D,\text{wake}} = 0.376743,$$

is only modestly above that simple pegged range.

The geodesic-pressure value,

$$C_{D,\text{geoP}} = 0.407006,$$

is still high, but this is consistent with the known pressure and divergence limitations of the current Android-local run.

Final Corrected Statement

The Harding pegged Ahmed-body run is not directly comparable to a clean Ahmed-body drag coefficient. Once peg drag and underbody interference are included,

$$C_D \approx 0.38\text{--}0.41$$

becomes plausible as a diagnostic result.

The method remains ERCOFTAC-relevant, but it is not yet ERCOFTAC-validated.

The correct next benchmark is an exact pegged or stilted Ahmed geometry with matched reference area, Reynolds number, ground clearance, support diameter, support height, and body slant angle.

The validation path should compare

$$C_D, \quad C_L, \quad C_p, \quad u/U_\infty, \quad \text{wake length,} \quad \text{recirculation center,} \quad W(\mathbf{x}), \quad C_{D,\text{geoP}}.$$

The strongest current conclusion is that the geodesic sectional boundary method with morphed normal shell diagnostics produces a plausible pegged-body drag and wake result, but it still requires matched workstation validation against an exact pegged ERCOFTAC-style geometry.